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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

0160606908

CHEMISTRY **5070/43**

Paper 4 Alternative to Practical **May/June 2010**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Write your answers in the spaces provided in the Question Paper.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

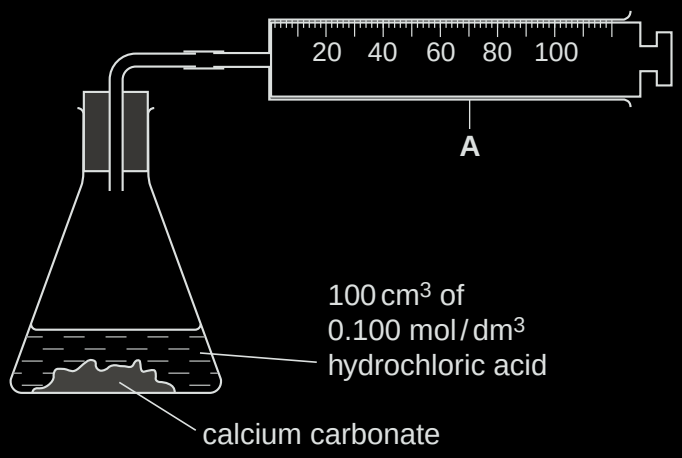
For Examiner's Use

This document consists of **14** printed pages and **2** blank pages.

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UNIVERSITY of CAMBRIDGE International Examinations

[Turn over



(a) Name the apparatus labelled **A**.

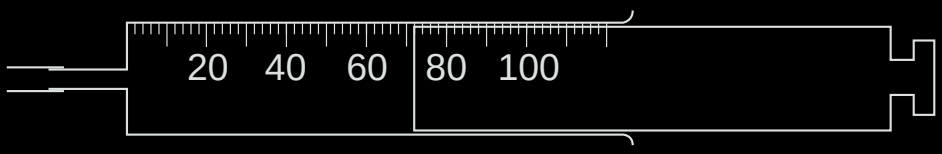
..... [1]

(b) Give a test to confirm the presence of carbon dioxide.

test .....

observation ..... [1]

(c) The diagram below shows apparatus **A** at the completion of the reaction.



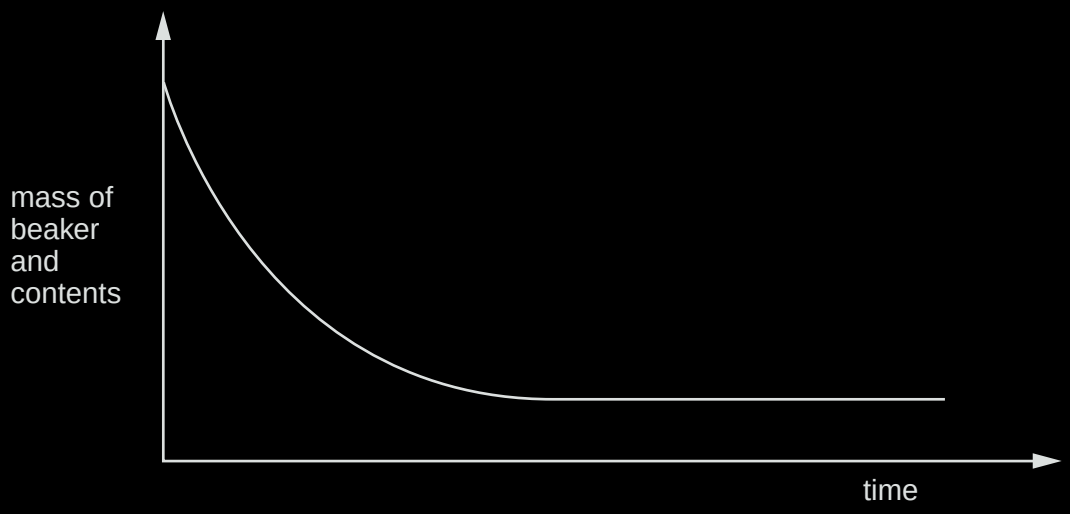
apparatus **A**

What volume of carbon dioxide was collected?

..... cm^3 [1]

(d) Using your answer to **(c)**, calculate the number of moles of carbon dioxide produced in the reaction.
[One mole of a gas occupies 24 000 cm^3 at room temperature and pressure.]

..... moles [1]



Which two solutions would give this graph?

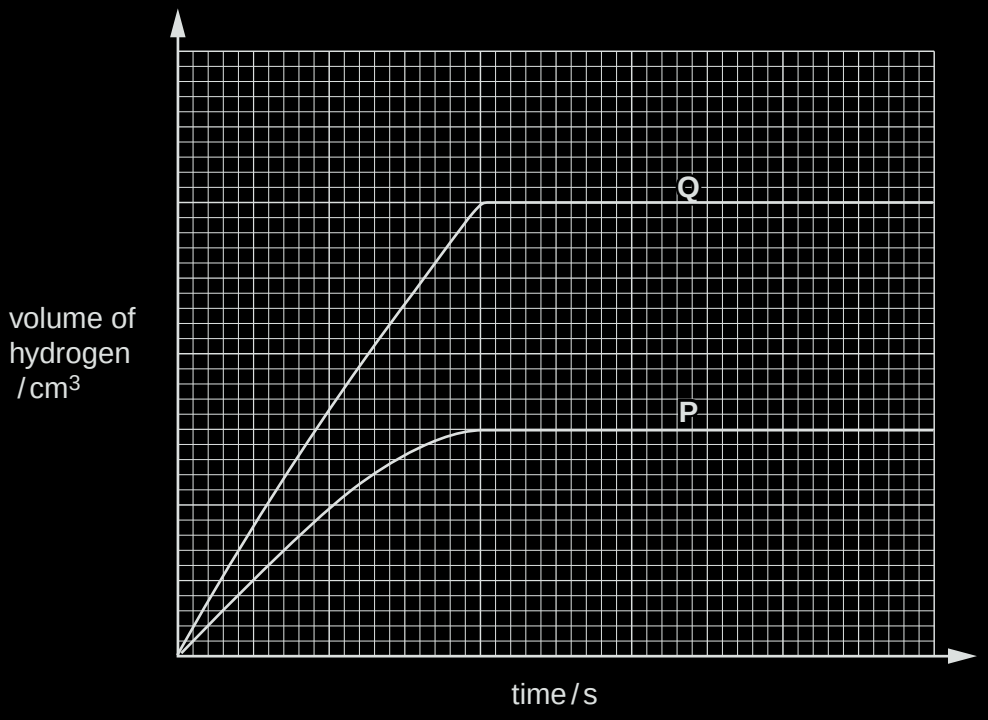
(a) aqueous copper(II) sulfate and aqueous ammonia ☐

(b) aqueous sodium carbonate and dilute nitric acid ☐

(c) aqueous sodium hydroxide and aqueous zinc sulfate ☐

(d) dilute hydrochloric acid and aqueous sodium sulfate ☐ [1]

[Total: 1]



Which acid solution would give the graph for experiment **Q**?

(a) 10 cm^3 of 0.050 mol/dm^3 hydrochloric acid

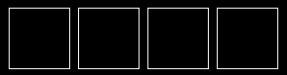
(b) 10 cm^3 of 0.200 mol/dm^3 hydrochloric acid

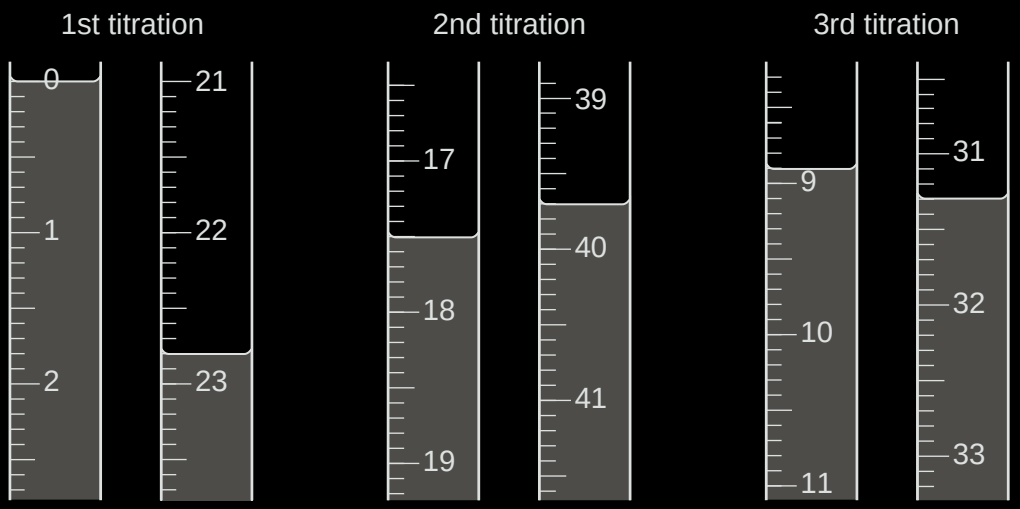
(c) 20 cm^3 of 0.100 mol/dm^3 hydrochloric acid

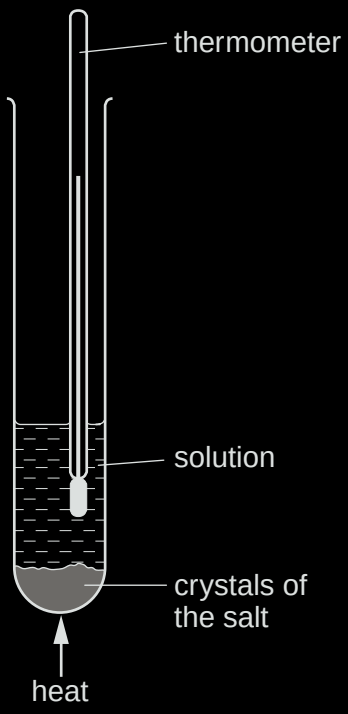
(d) 20 cm^3 of 0.200 mol/dm^3 hydrochloric acid [1]

[Total: 1]



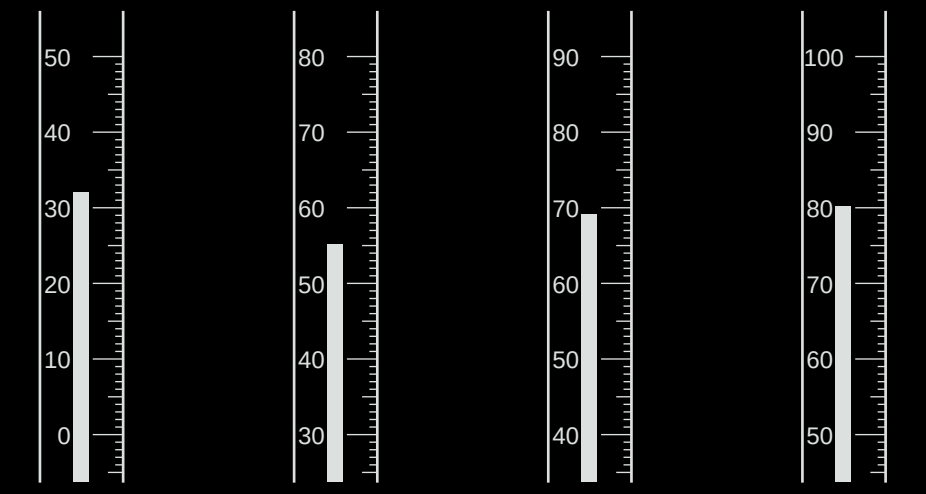



1st titration
0
1
2
21
22
23
2nd titration
17
18
19
39
40
41
3rd titration
9
10
11
31
32
33



10 g of water was transferred into a boiling-tube. To this 0.5 g of potassium chlorate(V) was added.
The tube and its contents were heated until all the solid dissolved. The tube was allowed to cool.
At the first sign of solid reappearing the temperature was noted.
The experiment was repeated using 1.0, 2.0, 3.0 and 4.0 g of potassium chlorate(V).

The diagrams below show parts of the thermometer stems giving the temperature at which the solid appeared.

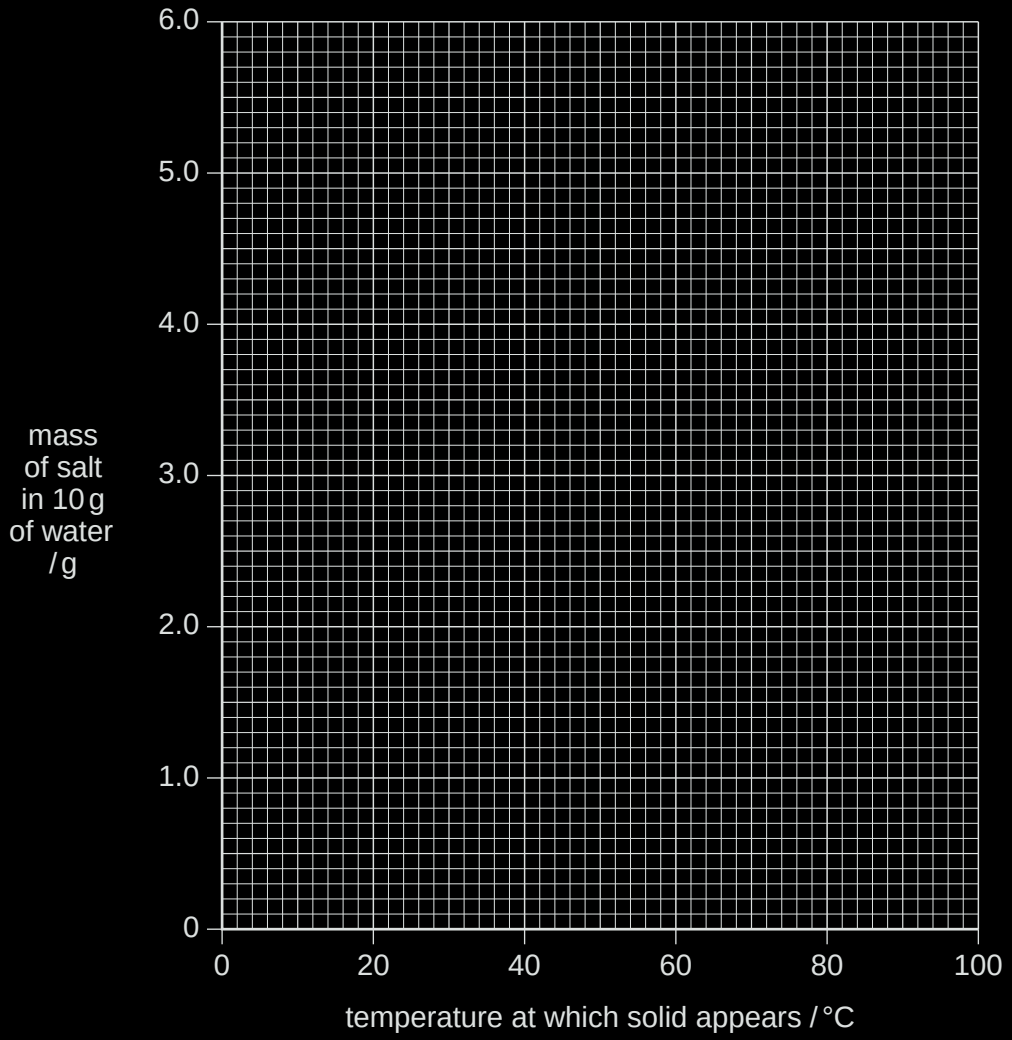


1.0 g potassium chlorate(V)

2.0 g potassium chlorate(V)

3.0 g potassium chlorate(V)

4.0 g potassium chlorate(V)


6.0
5.0
4.0
3.0
2.0
1.0
0
mass of salt in 10 g of water / g
0
20
40
60
80
100
temperature at which solid appears / °C

[4]